United States Patent Application Publication 20250276850
Kind Code A1
Publication Date September 04, 2025
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Load Handling System for Storage Racking

Abstract

A load handling system for storage racking includes a first bracket assembly for coupling a first horizontal guide rail to the storage racking, featuring a first racking column adapter with a column connection portion for attachment to a racking column and a support member extending perpendicularly. A first horizontal guide rail support device is coupled to the support member to secure the guide rail. A second bracket assembly similarly couples a second horizontal guide rail, with a second racking column adapter and a second guide rail support device. A vertical guide rail assembly, coupled to both horizontal guide rails, is horizontally movable along them via a first motor. A load handling platform, attached to the vertical guide rail assembly, is vertically movable along the vertical guide rail assembly by a second motor, facilitating efficient load handling within the storage racking system.

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Family ID: **95341242**

Appl. No.: **19/211023**

Filed: **May 16, 2025**

Related U.S. Application Data

parent US division 18917863 20241016 PENDING child US 19211023
us-provisional-application US 63590702 20231016
us-provisional-application US 63563970 20240312

Publication Classification

Int. Cl.: **B65G1/04** (20060101)

U.S. Cl.:

CPC **B65G1/0435** (20130101); B65G2203/0233 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 18/917,863, filed Oct. 16, 2024, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/590,702, filed Oct. 16, 2023 and titled “Modularized Rack Storage System” and U.S. Provisional Patent Application No. 63/563,970, filed Mar. 12, 2024 and titled “Modularized Rack Storage System”, each of which is incorporated herein by reference in its entirety.

BACKGROUND
[0002] At present, the logistics industry is developing rapidly, and the degree of automation is becoming increasingly important and advanced. Coupled with the increase in the frequency of various e-commerce promotional activities, it is difficult for the current warehousing situation to remain in a relatively stable state like the traditional business era, and it is necessary to frequently adjust the warehousing scale according to the number of orders generated by e-commerce promotions. At the same time, customers' needs for warehousing are also different. Some customers are cost-oriented and willing to sacrifice some warehousing efficiency, while some customers are efficiency-oriented and willing to relax cost requirements. At the same time, due to the different storage spaces of customers, it is difficult to unify the size specifications of all storage shelves. If customized for each customer's needs, although the needs of most customers can be solved, the high cost of customization and molding are unacceptable to logistics and warehousing suppliers.
SUMMARY
[0003] In some aspects, the techniques described herein relate to a load handling system for storage racking, the load handling system including: a first bracket assembly configured to couple a first horizontal guide rail with the storage racking, wherein the first bracket assembly includes: a first racking column adapter having a first column connection portion, wherein the first column connection portion is configured to attach to a column of the storage racking, and a first support member extending substantially perpendicularly from the first column connection portion; and a first horizontal guide rail support device coupled with the first support member, wherein the first horizontal guide rail support device is configured to secure the first horizontal guide rail; a second bracket assembly configured to couple a second horizontal guide rail with the storage racking, wherein the second bracket assembly includes: a second racking column adapter configured to attach to the column of the storage racking; and a second horizontal guide rail support device coupled with the second racking column adapter, wherein the second horizontal guide rail support device is configured to secure the second horizontal guide rail; a vertical guide rail assembly coupled with the first horizontal guide rail and the second horizontal guide rail, wherein the vertical guide rail assembly is configured to be moved horizontally along the first horizontal guide rail and the second horizontal guide rail by a first motor; and a load handling platform coupled with the vertical guide rail assembly, wherein the load handling platform is configured to be moved vertically along the vertical guide rail assembly by a second motor.
[0004] In some aspects, the techniques described herein relate to a load handling system, wherein the first horizontal guide rail support device includes: a guide rail support plate coupled with the first support member; and a plurality of pressure plates coupled with the guide rail support plate; wherein the first horizontal guide rail is configured to be secured between the guide rail support plate and the plurality of pressure plates.

[0005] In some aspects, the techniques described herein relate to a load handling system, wherein the guide rail support plate is further configured to support a first gear rack, wherein a first pinion gear coupled with the first motor is configured to interface with the first gear rack to move the vertical guide rail assembly horizontally along the first horizontal guide rail.
[0006] In some aspects, the techniques described herein relate to a load handling system, wherein the first bracket assembly is configured to enable adjustment of an angle of the first support member with respect to the column of the storage racking to account for variability in columns of the storage racking.
[0007] In some aspects, the techniques described herein relate to a load handling system, wherein the second bracket assembly is configured to enable adjustment of an angle of the second horizontal guide rail support device with respect to the column of the storage racking to account for variability in columns of the storage racking.
[0008] In some aspects, the techniques described herein relate to a load handling system, wherein the first bracket assembly is configured to enable adjustment of a distance between the column of the storage racking and the first horizontal guide rail support device to account for variability in columns of the storage racking.
[0009] In some aspects, the techniques described herein relate to a load handling system, wherein the second bracket assembly is configured to enable adjustment of a distance between the column of the storage racking and the second horizontal guide rail support device to account for variability in columns of the storage racking.
[0010] In some aspects, the techniques described herein relate to a load handling system for storage racking, the load handling system including: a plurality of adjustable bracket assemblies coupled with a column of the storage racking, each adjustable bracket assembly of the plurality of adjustable bracket assemblies including: a column adapter extending substantially perpendicularly from the column of the storage racking; and a horizontal guide rail support coupled with the column adapter; a plurality of horizontal guide rails coupled with the plurality of horizontal guide rail supports; a vertical guide rail assembly slidably coupled with the plurality of horizontal guide rails, wherein the vertical guide rail assembly is motorized to move horizontally along the plurality of horizontal guide rails; a load handling platform slidably coupled with the vertical guide rail assembly, wherein the load handling platform is motorized to move vertically along the vertical guide rail assembly; and a load handling device, wherein the load handling platform is configured to deliver the load handling device to manipulate a load located in the storage racking.
[0011] In some aspects, the techniques described herein relate to a load handling system, wherein the plurality of adjustable bracket assemblies are configured to enable adjustment of yaw, pitch, and roll of the plurality of adjustable bracket assemblies such that the plurality of horizontal guide rails are level and straight regardless of variability in column position of the storage racking.
[0012] In some aspects, the techniques described herein relate to a load handling system, wherein the plurality of adjustable bracket assemblies include: an adjustable upper bracket assembly coupled with the column of the storage racking; and an adjustable lower bracket assembly coupled with the column of the storage racking.
[0013] In some aspects, the techniques described herein relate to a load handling system, wherein the adjustable upper bracket assembly includes: an adjustable column adapter including: a column connection portion, wherein the column connection portion is configured to adjustably attach to the column of the storage racking; and a support member extending substantially perpendicularly from the column connection portion; and a horizontal guide rail support device coupled with the support member, wherein the horizontal guide rail support device is configured to adjustably secure a horizontal guide rail.
[0014] In some aspects, the techniques described herein relate to a load handling system, wherein the horizontal guide rail support device includes: a guide rail support plate coupled with the support member; and a plurality of pressure plates coupled with the guide rail support plate, wherein the

horizontal guide rail is configured to be secured between the guide rail support plate and the plurality of pressure plates.

[0015] In some aspects, the techniques described herein relate to a load handling system, wherein the guide rail support plate is further configured to support a gear rack, wherein the gear rack is configured to interface with a motor of the vertical guide rail assembly to move the vertical guide rail assembly horizontally along the horizontal guide rail.

[0016] In some aspects, the techniques described herein relate to a load handling system, wherein the adjustable lower bracket assembly includes: an adjustable column adapter configured to adjustably attach to the column of the storage racking; and an adjustable horizontal guide rail support coupled with a distal end of the adjustable column adapter, wherein the horizontal guide rail support is configured to support a horizontal guide rail.

[0017] In some aspects, the techniques described herein relate to a load handling system, wherein the horizontal guide rail includes a gear rack, wherein the gear rack is configured to interface with a motor of the vertical guide rail assembly to move the vertical guide rail assembly horizontally along the horizontal guide rail.

[0018] In some aspects, the techniques described herein relate to a load handling system for storage racking, the load handling system including: means for adjustably coupling an upper horizontal guide rail with the storage racking; means for adjustably coupling a lower horizontal guide rail with the storage racking; a vertical guide rail assembly slidably coupled with the upper horizontal guide rail and the lower horizontal guide rail, wherein the vertical guide rail assembly is motorized to move horizontally along the upper horizontal guide rail and the lower horizontal guide rail; a load handling platform slidably coupled with the vertical guide rail assembly, wherein the load handling platform is motorized to move vertically along the vertical guide rail assembly; and load handling means, wherein the load handling platform is configured to deliver the load handling means to manipulate a load located in the storage racking.

[0019] In some aspects, the techniques described herein relate to a load handling system, wherein the means for adjustably coupling the upper horizontal guide rail with the storage racking is configured to enable adjustment of pitch, yaw, and roll such that the upper horizontal guide rail are level and straight regardless of variability in column position of the storage racking.

[0020] In some aspects, the techniques described herein relate to a load handling system, wherein the means for adjustably coupling the lower horizontal guide rail with the storage racking is configured to enable adjustment of pitch, yaw, and roll such that the lower horizontal guide rail are level and straight regardless of variability in column position of the storage racking.

[0021] In some aspects, the techniques described herein relate to a load handling system, wherein the means for adjustably coupling the upper horizontal guide rail with the storage racking is configured to enable adjustment of a distance between the storage racking and the upper horizontal guide rail to account for variability in columns of the storage racking.

[0022] In some aspects, the techniques described herein relate to a load handling system, wherein the means for adjustably coupling the lower horizontal guide rail with the storage racking is configured to enable adjustment of a distance between the storage racking and the lower horizontal guide rail to account for variability in columns of the storage racking.

[0023] However, this list of features and advantages is not all-inclusive and many additional features and advantages are within the scope of the present disclosure. Moreover, it should be noted that the language used in the present disclosure has been principally selected for readability and instructional purposes, and not to limit the scope of the subject matter disclosed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] FIG. **1** depicts an example load handling system attached to storage racking.

[0025] FIG. **2** depicts an upper portion of an example load handling system attached to storage racking.

[0026] FIG. **3** is an end view of an upper portion of an example load handling system attached to storage racking.

[0027] FIG. **4** depicts components of an upper portion of a vertical guide rail assembly in an example load handling system.

[0028] FIGS. **5** depicts an example of an upper bracket assembly for attaching a load handling system to storage racking.

[0029] FIG. **6** is a top view of upper bracket assembly and depicts the angular adjustability of the upper bracket assembly in the horizontal direction parallel to the face of storage racking.

[0030] FIG. **7** is a side view of upper bracket assembly and depicts the angular adjustability of the upper bracket assembly in the vertical direction.

[0031] FIG. **8** depicts the linear adjustability of the upper bracket assembly in the vertical direction.

[0032] FIG. **9** depicts a lower portion of an example load handling system attached to storage racking.

[0033] FIG. **10** depicts an example of a lower bracket assembly for attaching a load handling system to storage racking.

[0034] FIG. **11** depicts another view of an example of a lower bracket assembly for attaching a load handling system to storage racking.

[0035] FIG. **12** depicts an example of a flexible motor mount for use in a load handling system.

[0036] FIG. **13** depicts an example of a flexible motor mount for use in a load handling system attached to the vertical guide rail assembly.

[0037] FIG. **14** depicts an example load handling platform and load handling device for use in a load handling system.

[0038] FIG. **15** depicts an example vertical slider assembly configured to move a load handling platform vertically along a vertical rail.

[0039] FIG. **16** is a cutaway view from the top of an example vertical slider assembly.

[0040] FIG. **17** depicts an example load handling device for use in a load handling system.

[0041] FIG. **18** depicts an overhead view of an example vertical guide rail assembly mounted between two rows of storage racking.

[0042] FIG. **19** depicts an overhead view of an example vertical guide rail assembly mounted between two rows of storage racking.

[0043] FIG. **20** depicts an overhead view of an example vertical guide rail assembly mounted between two rows of storage racking.

[0044] FIG. **21** shows an example load handing device coupled with vertical guide rail assembly and extending into storage racking from a different perspective.

[0045] FIG. **22** is a line drawing of an example workstation in a load handling system.

[0046] FIG. **23** depicts an example ground handling device for use in a load handling system.

[0047] FIG. **24** depicts an example elevated workstation with multiple pick up and drop off locations.

[0048] FIG. **25** depicts an example modularized storage system.

[0049] FIG. **26** is an illustration of an example load handling device.

[0050] FIG. **27** is a front view of an example load handling device.

[0051] FIG. **28** a perspective view of an example load handling device.

[0052] FIG. **29** depicts an example storage rack with corrugated decking.

[0053] FIG. **30** depicts another example embodiment of a modularized storage system.

[0054] FIG. **31** depicts an overhead view of another example embodiment of a modularized storage

system. system.
[0055] FIG. **32** is an overhead view of another embodiment of a modularized storage
DETAILED DESCRIPTION
[0056] The technology described in this disclosure relates to load handling in a warehouse or similar storage environment where vertical storage racking is employed and particularly to environments where dense storage solutions are employed. As an example, the technology provides for automated and efficient retrieval and replacement of loads in the storage racking in a dense, multi-deep storage system.
[0057] FIG. **1** depicts an example load handling system **100** attached to storage racking **101**. The load handling system **100** is designed to be universally compatible with a wide variety of storage racking systems. The load handling system **100** includes an upper horizontal guide rail **102**, a vertical guide rail assembly **104**, a lower horizontal guide rail **106**, a load handling platform **108**, and load handling device **114**. The upper horizontal guide rail **102** is connected to the storage racking **101** by upper bracket assembly **110**. The lower horizontal guide rail **106** is connected to the storage racking **101** by lower bracket assemblies **112**. The upper bracket assembly **110** and lower bracket assembly **112**, as described in more detail below, are designed to be adjustable in three dimensions to allow for variations in the storage racking **101** while maintaining the upper horizontal guide rail **102** and lower horizontal guide rail **106** in a straight and stable position to allow the vertical guide rail assembly **104** to travel freely along the upper horizontal guide rail **102** and the lower horizontal guide rail **106**.
[0058] The vertical guide rail assembly **104** connects to, and travels horizontally along, upper horizontal guide rail **102** and lower horizontal guide rail **106**. Load handling platform **108** connects to, and travels vertically along, the rails of the vertical guide rail assembly **104**. The horizontal movement of the vertical guide rail assembly **104** and the vertical movement of the load handling platform **108** allows the load handling device **114** to access and manipulate loads from any position in the storage racking **101**. While the loads depicted in the examples throughout the figures are totes, it should be noted that the load handling device **114** may be adapted to handle various types of loads or objects including, but not limited to, totes, trays, boxes, cartons, pallets, etc.
[0059] While the example of FIG. **1** depicts a single load handling platform **108** and load handling device **114** coupled with vertical guide rail assembly **104**, it should be appreciated that each vertical guide rail assembly **104** in load handling system **100** can accommodate multiple load handling platforms and/or load handling devices. For example, storage racking may be divided into two or more vertical zones, each with one or more load handling platforms and/or load handling devices deployed to service a vertical zone. In another embodiment, a first load handling device can remove and hold a first load from storage racking so a second load handling device on the same vertical guide rail assembly can access a second load that was stored behind the first load. It should be apparent that any number of load handling platforms and/or load handling devices can be used to access loads stored deeper in the storage racking.
[0060] FIG. **2** depicts an upper portion of an example load handling system **100** attached to storage racking **101**. As shown in the example of FIG. **2**, upper horizontal guide rail **102** may be an I or H type beam coupled with the storage racking **101** by the upper bracket assembly **110**. The vertical guide rail assembly **104** is slidably coupled with the upper horizontal guide rail **102** by trolley **202** and trolley **204** and is moved along the upper horizontal guide rail **102** by upper horizontal drive motor **206**. In some embodiments, multiple horizontal drive motors may attached to the vertical guide rail assembly **104** and move the vertical guide rail assembly **104** along the upper horizontal guide rail **102**. The additional horizontal drive motors allow for the vertical guide rail assembly to more quickly move loads and/or handle heavier loads.
[0061] In the example of FIG. **2**, the upper portion of the vertical guide rail assembly **104** includes a vertical guide rail mounting plates **208** which are bolted together and secure vertical rail **210** and vertical rail **212** of the vertical guide rail assembly **104** together. The other components of the upper

portion of the vertical guide rail assembly **104**, including, for example, upper horizontal drive motor **206**, trolley **202**, trolley **204**, and sliding power connector **214**, are also connected to vertical guide rail mounting plates **208**. Sliding power connector **214** interfaces with power rail **216** and provides power to the components of the vertical guide rail assembly **104** as it moves along the upper horizontal guide rail **102**.

[0062] As further shown in the example of FIG. **2**, the upper bracket assembly **110** supports power rail **216** and upper gear rack **218**. Upper gear rack **218** interfaces with a pinion gear (not shown) on upper horizontal drive motor **206**. The rack and pinion system drives the vertical guide rail assembly **104** horizontally along upper horizontal guide rail **102**.

[0063] FIG. **3** is an end view of an upper portion of an example load handling system **100** attached to storage racking **101**. The example of FIG. **3** shows upper bracket assembly **110** attached to storage racking **101**. The upper bracket assembly **110** supports the upper horizontal guide rail **102** via a guide rail support plate **302** and guide rail pressure plates **304**. Trolley **202** of the vertical guide rail assembly **104** rests on and moves along the upper horizontal guide rail **102**. As shown in example of FIG. **3**, upper gear rack **218** is coupled with the guide rail support plate **302** and extends along the front of the storage racking **101** parallel to upper horizontal guide rail **102**. Upper pinion gear **306** is coupled with upper horizontal drive motor **206** and engages with the upper gear rack **218** to drive the vertical guide rail assembly **104** along the upper horizontal guide rail **102**. As shown in the example of FIG. **3**, power rail **216** is coupled directly with upper bracket assembly **110** and engages with sliding power connector **214** to provide power to components of vertical guide rail assembly **104**. In other embodiments, guide rail support plate **302** is longer and power rail **216** is attached to the bottom of guide rail support plate **302**.

[0064] In some embodiments, to make installation of the load handling system easier, upper horizontal guide rail **102** is installed in multiple sections. To ensure proper alignment and strength at the joints of the multi-section upper horizontal guide rail **102**, splints **308** are secured over the joints of the upper horizontal guide rail **102**. When guide rail support plate **302** is installed on a joint in the upper horizontal guide rail **102**, the guide rail support plate **302** similarly helps with the alignment and rigidity of the upper horizontal guide rail **102**.

[0065] FIG. **4** depicts components of an upper portion of a vertical guide rail assembly **104** in an example load handling system **100**. The vertical guide rail assembly **104** includes vertical guide rails along which a load handling device travels vertically in the load handling system **100**. The example of FIG. **4** depicts a vertical guide rail **402** of a vertical guide rail assembly **104**. The vertical guide rail **402** is secured to the vertical guide rail mounting plates **208** via bracket **406** and bracket **404** creating stability in all directions when a load handling device is traveling along the vertical guide rail **402**. The vertical guide rail assembly **104** includes an anti-jump roller **408** that sits underneath the upper horizontal guide rail **102** and works with the wheels of trolley **202** to keep the vertical guide rail assembly **104** from bouncing off the upper horizontal guide rail **102**. In some embodiments, anti-jump roller **408** is mounted on a pivot plate that is supported by a spring such that the anti-jump roller **408** maintains contact with the underside of the upper horizontal guide rail **102** to decrease the chance that the vertical guide rail assembly **104** comes off the upper horizontal guide rail **102**.

[0066] FIGS. **5** depicts an example of an upper bracket assembly **110** for attaching a load handling system to storage racking. The upper bracket assembly **110** includes an upper racking column adapter **502**. A column adapter portion **503** of the upper racking column adapter **502** attaches to a column of storage racking **101** and a support member **504**, extending from the column adapter portion **503** of the upper racking column adapter **502**, supports the remaining components of the upper bracket assembly **110**. To support the upper horizontal guide rail **102**, guide rail support plate **302** is suspended from pressure plates **506** bolted together and compressed around the support member **504** of upper racking column adapter **502**. Guide rail pressure plates **304** secure the upper horizontal guide rail **102** to the guide rail support plate **302**. Guide rail support plate **302** also supports a rack plate **508** which houses upper gear rack **218** and power rail **216**.

[0067] Adjustability of the upper bracket assembly is important because storage racking is rarely perfectly level, plumb, and straight. Horizontal guide rails attached to the front of storage racking with typical vertical and horizontal variations would lead to horizontal guide rails that are similarly not level, plumb, or straight and a vertical guide rail assembly traveling along such horizontal guide rails would bind and/or jump the track leading to decreased reliability and wasted time. The techniques described herein address the deficiencies above by providing adjustability in multiple directions for upper horizontal guide rail **102** (as described with reference to FIG. **6** through FIG. **8**) and lower horizontal guide rail **106** (as described with reference to FIGS. **9** through FIG. **11**). This adjustability allows the upper horizontal guide rail **102** and lower horizontal guide rail **106** to be attached to storage racking **101** and run level, plumb, and straight, even if the storage racking **101** is not perfectly level, plumb, and/or straight, so that vertical guide rail assembly **104** can travel smoothly across the face of storage racking **101**.

[0068] FIG. **6** is a top view of upper bracket assembly **110** and depicts the ability to adjust the yaw of the upper bracket assembly **110** (i.e., horizontal angular adjustments left or right in the relative position between support member **504** and pressure plates **506** where the x-axis extends parallel to the face of storage racking **101** and the y-axis extends vertically). In one embodiment, pressure plates **506** allow for up to three degrees of adjustment around the support member **504** to correct for twist in the vertical posts of the storage racking **101**. In addition to the angular adjustability of upper bracket assembly **110**, pressure plates **506** can be positioned at any point along support member **504** to account for variations in depth of connection points to storage racking **101**.

[0069] FIG. **7** is a side view of upper bracket assembly **110** and depicts the ability to adjust the pitch of the upper bracket assembly **110** (i.e., vertical angular adjustments in the relative position between the distal end of support member **504** and the face of storage racking **101** where the x-axis is aligned parallel to the face of storage racking **101**). In one embodiment, column adapter portion **503** allows for up to three degrees of adjustment relative to a vertical post of storage racking **101** to account for unlevel storage racking **101**.

[0070] FIG. **8** depicts the linear adjustability of the upper bracket assembly **110** in the vertical direction and the ability to control the roll of the upper bracket assembly **110** (i.e., angular adjustments around the z-axis where the x-axis is aligned parallel to the face of storage racking **101** and the z-axis extends perpendicularly in the horizontal direction to the x-axis). While the upper racking column adapter **502** can be placed at multiple positions along a vertical post of storage racking **101**, the adjustability is somewhat limited due to the fact that bolt holes to fix the upper racking column adapter **502** are limited and at fixed intervals. To increase the vertical adjustability, the suspension bolts **802** that couple the guide rail support plate **302** with pressure plates **506** include additional threading so that fine adjustments to the vertical position of guide rail support plate **302** can be made so that upper horizontal guide rail **102** is level. Similarly, the vertical adjustability of suspension bolts **802** allow for adjustment in the roll of guide rail support plate **302** (e.g., by raising or lowering a pair of suspension bolts **802** on either side of support member **504**).

[0071] FIG. **9** depicts a lower portion of an example load handling system **100** attached to storage racking **101**. As depicted in the example of FIG. **9**, lower horizontal guide rail **106** may be a U or L shaped track in which guide wheels of the vertical guide rail assembly **104** can slidably connect with lower horizontal guide rail **106**. Lower horizontal guide rail **106** connects with the storage racking **101** by lower bracket assembly **112**.

[0072] In some embodiments a lower horizontal drive motor **902** works in tandem with upper horizontal drive motor **206** to move vertical guide rail assembly **104** along the upper horizontal guide rail **102** and lower horizontal guide rail **106**. In some embodiments, multiple lower horizontal drive motors may be attached to the vertical guide rail assembly **104** and move the vertical guide rail assembly **104** along the lower horizontal guide rail **106**. The additional horizontal drive motors allow for the vertical guide rail assembly **104** to more quickly move loads and/or handle heavier loads. In one embodiment, lower horizontal drive motor **902** may include a lower pinion gear (not shown) that engages with a lower gear rack **904** to drive vertical guide rail assembly **104** along lower horizontal guide rail **106**.

[0073] FIG. **10** depicts an example of a lower bracket assembly **112** for attaching a load handling system **100** to storage racking **101**. In one embodiment, lower bracket assembly **112** includes a lower racking column adapter **1004** and a lower horizontal rail support **1002**. As shown in the example of FIG. **10**, lower racking column adapter **1004** attaches to columns of storage racking **101** and lower horizontal rail support **1002** attaches to one end of lower racking column adapter **1004**. As described herein, adjustability of lower bracket assembly **112** allows lower horizontal guide rail **106** to be installed level, straight, and plumb with the upper horizontal guide rail, which allows the vertical guide rail assembly **104** to traverse the upper horizontal guide rail and lower horizontal guide rail smoothly without binding or jumping.

[0074] FIG. **11** depicts another view of an example of a lower bracket assembly **112** for attaching a load handling system **100** to storage racking **101**. As can be seen in the examples of FIG. **10** and FIG. **11**, lower racking column adapter **1004** includes a plurality of connection point holes **1102** that allow lower racking column adapter **1004** to be connected at various positions relative to the front of storage racking **101** so that lower horizontal guide rail **106** is straight and vertical guide rail assembly **104**, when attached to lower horizontal guide rail **106** and the upper horizontal guide rail, is plumb. Lower horizontal rail support **1002** provides additional vertical adjustability so that lower horizontal guide rail **106** is level when installed on storage racking **101**. In some embodiments, mounting slots **1104** in lower horizontal rail support **1002** and mounting slots **1106** in lower racking column adapter **1004** allow the lower horizontal rail support **1002** to be installed at various heights and angles relative to lower racking column adapter **1004** so that lower horizontal guide rail **106** can be installed level, plumb, and straight.

[0075] As described herein, existing storage racking may have defects due to wear and/or collision during installation or normal use. These defects create an imperfect foundation for the installation of horizontal guide rails used by the vertical guide rail assembly and leads to imperfect horizontal and/or vertical installation of the horizontal guide rails that the vertical guide rail assembly must cross. These imperfections include horizontal and vertical bending and/or deviation of the horizontal guide rails. When there is a large variation in the deviation between horizontal guide rails and pinion racks, a pinion gear on a fixed motor system engaging with the gear rack cannot accommodate the difference resulting in the vertical guide rail assembly getting stuck on the horizontal guide rails and unable to run.

[0076] FIG. **12** depicts an example of a flexible motor mount **1200** for use in a load handling system **100** to address the challenges described above. The flexible motor mount **1200** includes upper and lower limit guide wheels **1202** configured to keep the pinion gear **1204** aligned vertically with the gear rack. Front and rear limit guide wheels **1206** are configured to keep the pinion gear **1204** in contact with the gear rack. To minimize impact of variation in the horizontal and vertical position of the guide rails the flexible motor mount **1200** includes a compression spring assembly **1208** coupling motor mount **1210** with vertical guide rail assembly mount **1212**. The flexible motor mount **1200** thus allows the vertical guide rail assembly to smoothly traverse storage racking even if there are variations in the guide rails.

[0077] FIG. **13** depicts an example of another embodiment of a flexible motor mount **1300** for use in a load handling system **100** to address the challenges described above. Flexible motor mount **1300** includes front and rear limit guide wheels **1302** which are configured to keep the pinion gear **1308** in contact with the gear rack. To further minimize impact of variation in the horizontal and vertical position of the guide rails the flexible motor mount **1300** includes a first compression spring assembly **1304** and a second compression spring assembly **1306** coupling motor mount **1310** with a vertical guide rail assembly. First compression spring assembly **1304** provides horizontal flexibility and second compression spring assembly **1306** provides vertical flexibility for flexible motor mount **1300** and allows the vertical guide rail assembly to smoothly traverse storage racking even if there are variations in the guide rails.

[0078] FIG. **14** depicts an example load handling platform **108** and load handling device **114** for use in a load handling system **100**. Load handling platform **108** is slidably coupled with vertical rail **210** via vertical slider assembly **1402**, vertical rail **212**, and a second vertical slider assembly hidden from view in the example of FIG. **14**. The vertical slider assemblies include motors (not shown in the example of FIG. **14**) configured to move load handling platform **108** vertically along vertical rail **210** and vertical rail **212** and position load handling device **114** to manipulate loads (e.g., place, retrieve, move, etc.), such as tote **1404**, stored on storage racking **101**. To aid in manipulating loads, load handling device **114** includes retractable fingers **1406** which raise or lower to engage with a load being handled, as described in more detail elsewhere herein.

[0079] FIG. **15** depicts an example vertical slider assembly **1402** configured to move a load handling platform **108** vertically along vertical rail **210**. The components of vertical slider assembly **1402** are contained within or attached to assembly housing **1514**. Vertical drive motor **1502** is coupled with assembly housing **1514** and drives pinion gear **1506** which interfaces with gear rack **1504** to move **1402** and the attached load handling platform vertically along vertical rail **210**. Front limit guide wheel **1508** and **1510**, side limit guide wheel **1512** and **1516**, and rear limit guide wheels (not shown in the example of FIG. **15**) maintain the pinion gear **1506** in contact with gear rack **1504**.

[0080] FIG. **16** is a cutaway view from the top of an example vertical slider assembly **1402**. The example of FIG. **16** shows the profile of vertical rail **210** and how the components of vertical slider assembly **1402** engage with vertical rail **210** to move vertically along vertical rail **210**. Front limit guide wheel **1508** and **1510** roll along a front side of vertical rail **210** and rear limit guide wheel **1602** and **1604** roll along the rear side of a limit guide wheel flange of the vertical rail while side limit guide wheel **1512** and **1606** roll along opposite sides of vertical rail **210** to slidably couple vertical slider assembly **1402** with vertical rail **210** while maintaining pinion gear **1506** in contact with gear rack **1504**.

[0081] FIG. **17** depicts an example load handling device **114** for use in a load handling system **100**. In the example of FIG. **17**, load handling device **114** includes a body **1702** and fork **1704** and **1706**. Fork **1704** and **1706** include a plurality of fingers, such as finger **1708**, **1710**, **1712**, **1714**, **1716**, **1718**, **1720**, and **1722**. The fingers, all shown in a lowered position in the example of FIG. **17**, are configured to raise and lower in different configurations to manipulate a load. Manipulation of a load may include retrieving a load from storage racking or a pick-up and drop-off location and pulling it onto body **1702** or pushing a load from body **1702** onto the storage racking or a pick-up and drop-off location. In the embodiment shown in FIG. **17**, fork **1704** and **1706** are designed to reach loads that are stacked more than one layer deep on storage racking, as described with reference to FIGS. **18-20** below. In some embodiments, either one or both of fork **1704** and **1706**, in addition to extending either direction to manipulate loads on either side of load handling device **114**, may shift laterally along body **1702** to accommodate loads of varying width.

[0082] While the horizontal and vertical guide rails can be installed level, plumb, and straight according to the techniques described herein to optimize movement of load handling device **114** along storage racking **101**, variations in storage racking can lead to items on the storage racking shelves being located at various distances from load handling platform **108**. If those distances are not accounted for by load handling device **114**, attempts to retrieve a load from storage racking **101** may fail because the distance fork **1704** and **1706** extends could either be too short or too far to accurately retrieve the load. In one embodiment, to account for these variations in distance, load handling system **100** can measure the distance to each object with location information so that load handling device **114** can accurately retrieve the object. In one embodiment, each object stored on storage racking **101** may be marked with a barcode or some other indicator that can be read by a camera or other reader on load handling platform **108** or load handling device **114** so that a distance

can be determined based on the barcode.
[0083] FIGS. **18-20** depict an example process for manipulating loads on storage racking **101** by load handling device **114**. FIG. **18** depicts an overhead view of vertical guide rail assembly **104** mounted between two rows of storage racking **101**. According to the techniques described herein, when mounted to one of the rows of storage racking on horizontal and vertical guide rails, load handling device **114** can access any position along either side of storage racking. As shown in the example of FIG. **18**, load handling device **114** is extending fork **1704** and **1706** in the direction away from where vertical guide rail assembly **104** is attached to storage racking **101**. When manipulating a load, such as tote **1802**, load handling device **114** lowers finger **1708**, **1712**, **1716**, and **1720** to encompass the corners of tote **1802** and retrieve or store the tote **1802** on storage racking **101**. It should be appreciated, particularly in light of the following examples, that various combinations of fingers and fork depths can be used to access any position in storage racking.
[0084] FIG. **19** depicts an overhead view of vertical guide rail assembly **104** mounted between two rows of storage racking **101**. As shown in the example of FIG. **19**, load handling device **114** is extending fork **1704** and **1706** in the direction toward where vertical guide rail assembly **104** is attached to storage racking **101**. When manipulating a load, such as tote **1902**, load handling device **114** lowers finger **1710**, **1714**, **1718**, and **1722** to encompass the corners of tote **1902** and retrieve or store the tote **1902** from or to storage racking **101**.
[0085] Increasing the stroke length of fork **1704** and **1706** and using a different combination of fingers allows load handling device **114** to reach deeper into storage racking **101** and manipulate loads. Increasing the reach depth allows for fewer load handling systems to be deployed in a system than if each load handling device could only reach the first row of objects in storage racking. FIG. **20** depicts an overhead view of vertical guide rail assembly **104** mounted between two rows of storage racking **101**. As shown in the example of FIG. **20**, load handling device **114** is extending fork **1704** and **1706** beyond the first row of objects and accessing an object, for example tote **2002**, in the second row. To retrieve or store tote **2002** load handling device **114** lowers finger **1708**, **1712**, **1716**, and **1720** to encompass the corners of tote **2002** and retrieve or store the tote **2002** from or to storage racking **101**. With the configuration shown in the example of FIG. **20**, load handling device **114** can retrieve tote **2002** and **2004** simultaneously or individually.
[0086] FIG. **21** shows an example load handing device **114** coupled with vertical guide rail assembly **104** and extending into storage racking **101** from a different perspective.
[0087] In some embodiments, load handling system **100** may be part of a larger fulfillment environment where load handling system **100** delivers a load to human workers located at workstations throughout the facility. In one embodiment, a workstation may be incorporated into storage racking **101**. FIG. **22** depicts an example workstation integrated into storage racking **101**. In the example of FIG. **22**, a workstation may include one or more racks installed at an angle, such as rack **2204**. The angled rack provides easier access to the contents of a tote, such as tote **2206**, for a worker located at the workstation. A load handling system **100** as described herein may be attached to the opposite side of storage racking **101** from the work station. In the example of FIG. **22**, load handling device **114** of load handling system **100** traverses storage racking **101** and picks up and drops off totes to the workstation.
[0088] In some embodiments, workstations are located remotely from storage racking and loads, such as totes, must be moved from storage racking to a workstation. Various solutions, including automated ground vehicles, have been implemented to transport loads between storage racking and workstations. FIG. **23** depicts an example ground handling device, such as automated ground vehicle **2300**, for use as part of load handling system **100**. In a system such as load handling system **100** where automated ground vehicle **2300** may be interacting with loads at varying heights it would be beneficial to have an automated ground vehicle that has an adjustable load interface **2302**. Lifting mechanism **2304** adjusts the height of load interface **2302** depending on the context of the interaction by automated ground vehicle **2300**. For example, to allow for easier maneuverability of

automated ground vehicle **2300** throughout storage racking, pick up and drop off locations, as well as shelves and vertical guide rail assembly **104** may be elevated off of the floor. To retrieve a load, automated ground vehicle **2300** raises adjustable load interface **2302** and lifts the load off of an elevated pick up and drop off location in storage racking. Once the load has been retrieved, adjustable load interface **2302** is retracted and automated ground vehicle **2300** can navigate throughout the facility without interfering with overhead objects.

[0089] FIG. **24**, depicts an example elevated workstation **2400** with multiple pick up and drop off locations **2402**. Automated ground vehicle **2300** transfers a load between storage racking and elevated workstation **2400**. As depicted in the example of FIG. **24**, automated ground vehicle **2300** is retrieving tote **2404** from pick up and drop off location **2402**. Automated ground vehicle **2300** drives under pick up and drop off location **2402** and raises adjustable load interface **2302** to lift tote **2404** off of elevated workstation **2400**. After lifting tote **2404**, automated ground vehicle **2300** moves away and can lower adjustable load interface **2302** to avoid interference with other objects in the facility.

[0090] In some embodiments, instead of automated ground vehicle **2300** using adjustable load interface **2302** to lift tote **2404** to an ergonomic height, a component of the pick up and drop off location **2402** may be configured to raise and lower the tote. For example, in one embodiment, automated ground vehicle **2300** may be configured to navigate on a closed loop to a load transition station where load handling device **114** deposits a load on automated ground vehicle **2300**. Automated ground vehicle **2300** carries the load on the closed loop to a workstation where a lifting platform raises the load off of automated ground vehicle **2300** up to an ergonomic height for a worker at the workstation. In some embodiments, the workstation may include a conveyor to move one or more loads from the lifting platform to a worker and from the worker to a lowering platform where the lowering platform places the load on a automated ground vehicle **2300** to return on the closed loop to a transition station where load handling device **114** retrieves the load from automated ground vehicle **2300**.

[0091] FIG. **25** depicts a modularized storage system **2500**, which can adapt to the needs and storage space constraints of various environments, while maintaining a reasonable production cost. To achieve low-cost flexibility, the modularized storage system **2500** in the example of FIG. **25** utilizes standard storage racking **101** but may be configured to work with any number of storage racking configurations. Modularized storage system **2500** includes a load handling system **100** as described herein and one embodiment is depicted in the example of FIG. **25**. Horizontal guide rails, such as upper horizontal guide rail **102** and **106**, are attached to the storage racking **101**. In the example of FIG. **25**, vertical guide rail assembly **104** includes load handling platform **108** and load handling device **2514**. In the example of FIG. **25**, load handling device **2514** is configured to drive off of the load handling platform **108** and retrieve objects, such as tote **2516**, from storage racking **101**.

[0092] As described herein, load handling platform **108** is configured to move vertically along vertical guide rail assembly **104** and deliver the load handling device **2514** to any shelf of the storage racking **101**. Similarly, the vertical guide rail assembly **104** is configured to move horizontally along the storage racking **101** to deliver load handling device **2514** to any point in storage racking **101**. In some embodiments, the load handling platform **108** may further include imaging or other sensing devices to orient itself and deliver load handling device **2514** to the appropriate location on the storage racking **101**.

[0093] As shown in the example of FIG. **25**, the modularized storage system **2500** includes load transition stations, such as load transition station **1522**. The load transition stations allow for the modularized storage system **2500** to use a multi-deep dense storage system (i.e., multiple objects can be stored on a single row). Load handling device **2514** may use the load transition station **1522** to hold an object while returning to retrieve the next object in a row. Additionally, the load transition station **1522** may be used as a pick-up and drop-off area for automated ground vehicles, such as automated ground vehicle **2300**, to retrieve objects and transport them to other areas of the storage facility for further processing. In some embodiments, automated ground vehicle **2300** may be equipped with wireless charging equipment that interacts with a wireless charging station when stopped at a load transition station.

[0094] FIG. **26** is an illustration of an example load handling device **2514**. The load handling device **2514** may include drive wheel **2602** and **2604**. The drive wheels allow the load handling device **2514** to detach from load handling platform **108** and navigate the storage rack **101** independently of the vertical guide rail assembly **104**. In some embodiments, drive wheels **2602** and **2604** can operate independently allowing the load handling device **2514** to travel omnidirectionally.

[0095] As shown in the example of FIG. **26**, the Load handling device **114** may include one or more forks **2606** that are configured to handle objects. In some embodiments, the forks **2606** are configured to handle multiple objects at once, effectively reducing the number of trips load handling device **2514** needs to make to get to deeper objects in the multi-deep storage system. The forks **2606** may further include one or more guide wheels **2608** that are configured to support the forks and assist the load handling device **2514** in navigating the storage rack **101**. For example, the guide wheels **2608** and the drive wheels **2602** and **2604** may be configured to interface with tracks in corrugated decking of the storage rack **101** that guide the load handling device **2514** in and out of the storage rack **101**. While the example of FIG. **26** depicts load handling device **2514** having two forks, it should be understood that any number of forks may be present on a load handling device **2514** to accommodate different types of loads.

[0096] FIG. **27** is a front view of an example load handling device **2514**. Load handling device **2514** includes sensors to help navigate a storage rack. For example, load handling device **2514** may include object detection sensor **2710** and **2712**. Object detection sensor **2710** and **2712** may be configured to detect proximity to and position of objects with which load handling device **2514** may interact. In one embodiment, object detection sensors **2710** and **2712** may determine if an object is in a suitable position for load handling device **2514** to lift and retrieve. In some embodiments, object detection sensor **2710** and **2712** can detect the beginning and end of an object to identify whether the object is positioned within the lifting zone of load handling device **2514**.

[0097] Load handling device **2514** may further include one or more device position sensors, such as device position sensor **2714**. Device position sensor **2714** may be configured to determine the position of load handling device **2514** within a storage rack. For example, device position sensor **2714** may be configured to count marks, holes, or other position identifying features on the corrugated decking or work bays of a storage rack to determine its position. Additionally, information from device position sensors and motor encoders for the motors operating drive wheel **2602** and **2604** may be combined to better determine a position of load handling device **2514**. In some embodiments, storage racking may include reference points (e.g., the front of the rack, a specific mark or identifier within a rack) to signify a position within the rack to load handling device **2514**.

[0098] FIG. **28** is another view of an example load handling device **2514**. As seen in the example of FIG. **26**, the forks **2606** of load handling device **2514** may include one or more spring loaded idler wheels **2816** to maintain the position of load handling device **2514** in the tracks of the corrugated decking. Additionally, the forks of load handling device **2514** may include object retention devices, such as object retention device **2818**, that provide a retention mechanism for objects being transported by load handling device **2514**.

[0099] FIG. **29** illustrates an example storage rack **101** with corrugated decking **2902**. It should be noted that the modularized storage system described herein is configured to work with conventional storage racking systems and there is no need for custom rack storage, thus reducing cost to the storage facility. In some embodiments, as shown in the example of FIG. **29**, the corrugated decking may be realized by installing a plurality of tracks, such as track **2902**, at specified intervals to accommodate the wheels of a handling vehicle. Additionally, using tracks to create the corrugated decking allows for spaces, such as space **2904**, between the tracks. These spaces allow vision systems included in the load handling platform **108** and/or the load handling device **2514** to identify objects stored within storage racking **101**. For example, QR codes, barcodes, or other identifying marks on objects in storage racking **101** may be aligned with the spaces such that vision systems of the load handling platform **108** and/or load handling device **2514** may identify the objects. In other embodiments, load handling platform **108** and/or load handling device **2514** may use image processing to identify objects and their locations without identifying marks.

[0100] FIG. **30** illustrates another embodiment of a modularized storage system **3000**. In the example of FIG. **30**, the modularized storage system **3000** includes work bays, such as work bay **3002**, where multiple load handling devices, such as load handling device **2514**, may work to stage and retrieve objects from storage racking **101**. The addition of work bays in the modularized storage system **3000** allows the handling vehicles to shuffle objects within storage racking **101** without having to utilize the load handling platform **108** to navigate off of the current level of storage racking **101** to access objects deeper in storage racking **101**. This configuration provides additional efficiencies in accessing objects in storage racking **101**. As shown in the example of FIG. **30**, upper horizontal guide rail **102** and lower horizontal guide rail **106** may be mounted on the storage rack **101** adjacent to the work bays such that vertical guide rail assembly **104** and load handling platform **108** may access the work bays. Work bays may include charging stations, such as charging station **3004**, where load handling device **2514** can charge onboard batteries.

[0101] FIG. **31** is an overhead view of another embodiment of a modularized storage system **3100**. In the example of FIG. **31**, work bays **3102** may be located on all sides of storage racking **101** and load handling devices can access objects from any direction within the storage rack. In one embodiment, horizontal and vertical rails with vehicle platforms may be installed on one or more sides of storage racking **101**. However, it may be cost prohibitive to have multiple vehicle platforms servicing a single storage cube. In other embodiments, horizontal guide rails may be installed on all sides of the storage rack and detachable, rotating corner sections **3104** provide for transitions of the vertical guide rail assembly from side to side of the storage rack.

[0102] FIG. **32** is an overhead view of another embodiment of a modularized storage system **3200**. In the example of FIG. **32**, multiple storage racks, for example storage rack **3201** and **3203**, may be serviced by a single load handling platform **108** and vertical guide rail assembly **104** coupled with storage rack **3201**. Load handling platform **108** may include a retractable ramp **3230** configured to allow load handling device **2514** to drive from load handling platform **108** onto a storage rack. While a single retractable ramp **3230** is shown in the example of FIG. **32**, it should be understood that a split platform (one for each wheel of load handling device **2514**) may be employed. With the retractable ramp **3230** retracted, load handling platform **108** is free to rotate in all directions to access storage rack **3203**. For example, load handling platform **108** may rotate 180 degrees and extend retractable ramp **3230** so that load handling device **2514** may access storage rack **3203**.

[0103] In some embodiments, load handling platform **108** may include a load handling device **114**, such as the handling device described above, in addition to being a platform to carry load handling device **2514**. Load handling device **114** may be configured to manipulate objects, such as objects **3232** and **3234**, independent of load handling device **2514**.

[0104] As described above storage racks may include a work bay, such as work bay **3236**, to allow a handling vehicle to move objects around the storage rack. The handling vehicles described herein are configured to travel on a flat surface and the wheels can operate independently so that the handling vehicle can travel in any direction in a work bay as well as navigate in the tracks of the corrugated decking as described above.

[0105] The foregoing description, for purpose of explanation, has been described with reference to various embodiments and examples. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The various embodiments and examples were chosen and described in order to best explain the principles of the invention and its practical applications, to thereby enable others skilled in the art to utilize the innovative technology with various modifications as may be suited to the particular use contemplated. For instance, it should be understood that the technology described herein can be practiced without these specific details in some cases. Further, various systems, devices, and structures are shown in block diagram form in order to avoid obscuring the description. For instance, various implementations are described as having particular hardware, software, and user interfaces.

[0106] In some instances, various implementations may be presented herein in terms of algorithms and symbolic representations of operations on data bits within a computer memory. An algorithm is here, and generally, conceived to be a self-consistent set of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0107] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the following discussion, it is appreciated that throughout this disclosure, discussions utilizing terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” or the like, refer to the action and methods of a computer system that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0108] A data processing system suitable for storing and/or executing program code, such as the computing system and/or devices discussed herein, may include at least one processor coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories that provide temporary storage of at least some program code in order to reduce the number of times code must be retrieved from bulk storage during execution. Input or I/O devices can be coupled to the system either directly or through intervening I/O controllers. The data processing system may include an apparatus may be specially constructed for the required purposes, or it may comprise a general-purpose computer selectively activated or reconfigured by a computer program stored in the computer.

[0109] The foregoing description has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the specification to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the disclosure be limited not by this detailed description, but rather by the claims of this application. As will be understood by those familiar with the art, the specification may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. Likewise, the particular naming and division of the modules, routines, features, attributes, methodologies and other aspects may not be mandatory or significant, and the mechanisms that implement the specification or its features may have different names, divisions, and/or formats.

[0110] Furthermore, the modules, routines, features, attributes, methodologies and other aspects of the disclosure can be implemented as software, hardware, firmware, or any combination of the foregoing. The technology can also take the form of a computer program product accessible from a computer-usable or computer-readable medium providing program code for use by or in connection with a computer or any instruction execution system. Wherever a component, an example of which is a module or engine, of the specification is implemented as software, the component can be implemented as a standalone program, as part of a larger program, as a plurality of separate programs, as a statically or dynamically linked library, as a kernel loadable module, as firmware, as resident software, as microcode, as a device driver, and/or in every and any other way known now or in the future. Additionally, the disclosure is in no way limited to implementation in any specific programming language, or for any specific operating system or environment. Accordingly, the disclosure is intended to be illustrative, but not limiting, of the scope of the subject matter set forth in the following claims.

Claims

1. A load handling system for storage racking, the load handling system comprising: a first bracket assembly configured to couple a first horizontal guide rail with the storage racking, wherein the first bracket assembly comprises: a first racking column adapter having a first column connection portion, wherein the first column connection portion is configured to attach to a column of the storage racking, and a first support member extending substantially perpendicularly from the first column connection portion; and a first horizontal guide rail support device coupled with the first support member, wherein the first horizontal guide rail support device is configured to secure the first horizontal guide rail; a second bracket assembly configured to couple a second horizontal guide rail with the storage racking, wherein the second bracket assembly comprises: a second racking column adapter configured to attach to the column of the storage racking; and a second horizontal guide rail support device coupled with the second racking column adapter, wherein the second horizontal guide rail support device is configured to secure the second horizontal guide rail; a vertical guide rail assembly coupled with the first horizontal guide rail and the second horizontal guide rail, wherein the vertical guide rail assembly is configured to be moved horizontally along the first horizontal guide rail and the second horizontal guide rail by a first motor; and a load handling platform coupled with the vertical guide rail assembly, wherein the load handling platform is configured to be moved vertically along the vertical guide rail assembly by a second motor.
2. The load handling system of claim 1, wherein the first horizontal guide rail support device comprises: a guide rail support plate coupled with the first support member; and a plurality of pressure plates coupled with the guide rail support plate; wherein the first horizontal guide rail is configured to be secured between the guide rail support plate and the plurality of pressure plates.
3. The load handling system of claim 2, wherein the guide rail support plate is further configured to support a first gear rack, wherein a first pinion gear coupled with the first motor is configured to interface with the first gear rack to move the vertical guide rail assembly horizontally along the first horizontal guide rail.
4. The load handling system of claim 1, wherein the first bracket assembly is configured to enable adjustment of an angle of the first support member with respect to the column of the storage racking to account for variability in columns of the storage racking.
5. The load handling system of claim 1, wherein the second bracket assembly is configured to enable adjustment of an angle of the second horizontal guide rail support device with respect to the column of the storage racking to account for variability in columns of the storage racking.
6. The load handling system of claim 1, wherein the first bracket assembly is configured to enable adjustment of a distance between the column of the storage racking and the first horizontal guide rail support device to account for variability in columns of the storage racking.
7. The load handling system of claim 1, wherein the second bracket assembly is configured to enable adjustment of a distance between the column of the storage racking and the second horizontal guide rail support device to account for variability in columns of the storage racking.
8. A load handling system for storage racking, the load handling system comprising: a plurality of

adjustable bracket assemblies coupled with a column of the storage racking, each adjustable bracket assembly of the plurality of adjustable bracket assemblies comprising: a column adapter extending substantially perpendicularly from the column of the storage racking; and a horizontal guide rail support coupled with the column adapter; a plurality of horizontal guide rails coupled with the plurality of horizontal guide rail supports; a vertical guide rail assembly slidably coupled with the plurality of horizontal guide rails, wherein the vertical guide rail assembly is motorized to move horizontally along the plurality of horizontal guide rails; a load handling platform slidably coupled with the vertical guide rail assembly, wherein the load handling platform is motorized to move vertically along the vertical guide rail assembly; and a load handling device, wherein the load handling platform is configured to deliver the load handling device to manipulate a load located in the storage racking.

9. The load handling system of claim 8, wherein the plurality of adjustable bracket assemblies are configured to enable adjustment of yaw, pitch, and roll of the plurality of adjustable bracket assemblies such that the plurality of horizontal guide rails are level and straight regardless of variability in column position of the storage racking.

10. The load handling system of claim 8, wherein the plurality of adjustable bracket assemblies include: an adjustable upper bracket assembly coupled with the column of the storage racking; and an adjustable lower bracket assembly coupled with the column of the storage racking.

11. The load handling system of claim 10, wherein the adjustable upper bracket assembly comprises: an adjustable column adapter comprising: a column connection portion, wherein the column connection portion is configured to adjustably attach to the column of the storage racking; and a support member extending substantially perpendicularly from the column connection portion; and a horizontal guide rail support device coupled with the support member, wherein the horizontal guide rail support device is configured to adjustably secure a horizontal guide rail.

12. The load handling system of claim 11, wherein the horizontal guide rail support device comprises: a guide rail support plate coupled with the support member; and a plurality of pressure plates coupled with the guide rail support plate, wherein the horizontal guide rail is configured to be secured between the guide rail support plate and the plurality of pressure plates.

13. The load handling system of claim 12, wherein the guide rail support plate is further configured to support a gear rack, wherein the gear rack is configured to interface with a motor of the vertical guide rail assembly to move the vertical guide rail assembly horizontally along the horizontal guide rail.

14. The load handling system of claim 10, wherein the adjustable lower bracket assembly comprises: an adjustable column adapter configured to adjustably attach to the column of the storage racking; and an adjustable horizontal guide rail support coupled with a distal end of the adjustable column adapter, wherein the horizontal guide rail support is configured to support a horizontal guide rail.

15. The load handling system of claim 14, wherein the horizontal guide rail comprises a gear rack, wherein the gear rack is configured to interface with a motor of the vertical guide rail assembly to move the vertical guide rail assembly horizontally along the horizontal guide rail.

16. A load handling system for storage racking, the load handling system comprising: means for adjustably coupling an upper horizontal guide rail with the storage racking; means for adjustably coupling a lower horizontal guide rail with the storage racking; a vertical guide rail assembly slidably coupled with the upper horizontal guide rail and the lower horizontal guide rail, wherein the vertical guide rail assembly is motorized to move horizontally along the upper horizontal guide rail and the lower horizontal guide rail; a load handling platform slidably coupled with the vertical guide rail assembly, wherein the load handling platform is motorized to move vertically along the vertical guide rail assembly; and load handling means, wherein the load handling platform is configured to deliver the load handling means to manipulate a load located in the storage racking.

17. The load handling system of claim 16, wherein the means for adjustably coupling the upper horizontal guide rail with the storage racking is configured to enable adjustment of pitch, yaw, and roll such that the upper horizontal guide rail are level and straight regardless of variability in column position of the storage racking.

18. The load handling system of claim 16, wherein the means for adjustably coupling the lower horizontal guide rail with the storage racking is configured to enable adjustment of pitch, yaw, and roll such that the lower horizontal guide rail are level and straight regardless of variability in column position of the storage racking.

19. The load handling system of claim 16, wherein the means for adjustably coupling the upper horizontal guide rail with the storage racking is configured to enable adjustment of a distance between the storage racking and the upper horizontal guide rail to account for variability in columns of the storage racking.

20. The load handling system of claim 16, wherein the means for adjustably coupling the lower horizontal guide rail with the storage racking is configured to enable adjustment of a distance between the storage racking and the lower horizontal guide rail to account for variability in columns of the storage racking.
